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ToF-SIMS analysis of ultrathin films and their fragmentation patterns

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Note: This paper is part of the Special Topic Collection Celebrating the Achievements and Life of Joe Greene.

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ABSTRACT

Organic thin films are of great interest due to their intriguing interfacial and functional properties, especially for device applications such as thin-film transistors and sensors. As their thickness approaches single nanometer thickness, characterization and interpretation of the extracted data become increasingly complex. In this study, plasma polymerization is used to construct ultrathin films that range in thickness from 1 to 20 nm, and time-of-flight secondary ion mass spectrometry coupled with principal component analysis is used to investigate the effects of film thickness on the resulting spectra. We demonstrate that for these cross-linked plasma polymers, at these thicknesses, the observed trends are different from those obtained from thicker films with lower degrees of cross-linking: contributions from ambient carbon contamination start to dominate the mass spectrum; cluster-induced nonlinear enhancement in secondary ion yield is no longer observed; extent of fragmentation is higher due to confinement of the primary ion energy; and the size of the primary ion source also affects fragmentation (e.g., Bi₁ versus Bi₅). These differences illustrate that care must be taken in choosing the correct primary ion source as well as in interpreting the data.

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I. INTRODUCTION

Organic thin films are of great interest due to their extensive applications in the fields of microelectronics, optics, and interfacial devices such as sensors for medical and environmental monitoring.¹ In the case of sensors, the essence of the technology is in their ability to transduce a chemical/physical event into a digital electronic signal, and biological sensors, especially, require a thin interfacial layer between the electrical transducer (i.e., microelectronics component) and the surrounding biological environment. This is because both components are not compatible with each other: attachment of biomolecules such as proteins onto a non-natural surface leads to nonspecific adsorption, conformational changes, and/or loss of activities;^{1–3} and microelectronics, including both inorganic and organic semiconductors, are sensitive to contamination and oxidation.^{4–6} The ideal interfacial layer, therefore, would be thin enough to maximize sensitivity, yet sufficiently robust and

biocompatible to allow the device to be stable, reproducible, and reusable.

In these fields, polymeric films are usually employed for this task due to their ability to meet strict large-scale processing requirements that include ultrathin film designs (<10 nm), mechanical flexibility, and ability to cover large areas or confer to unusual topographies.⁸ They can also be chemically stable and designed to have specific functionalities such as biocompatibility.⁹ Various wet and dry processing techniques exist for their production, and of these, plasma polymerization is suited for its ability to produce pinhole-free, flat surface films that are chemically and mechanically stable and have consistent and wide-ranging thickness control.^{7,10–12} Plasma polymerization is a specific type of process in plasma chemistry, which involves reactions between plasma species or between plasma and surfaces,^{13,14} with the unique feature of being able to polymerize unconventional starting materials such as saturated alkanes through random decomposition and free-radical

02 September 2025 17:23:29

reactions.¹⁵ It produces an irregular three-dimensional crosslinked network, which confers desirable properties such as chemical inertness, insolubility, mechanical strength, and thermal stability.^{15–17}

One disadvantage is that the resulting chemistry of the film does not always represent the starting feed material as is typically the case for molecular films, and experimental conditions can greatly affect the functional groups on the surface of these crosslinked plasma films.^{18–21} However, even though the film chemistry may differ from the feed material, these thin films typically have a uniform and homogeneous chemical structure. One technique useful for characterizing the surface chemistry of thin films is time-of-flight secondary ion mass spectrometry (ToF-SIMS) due to its nanometer surface sensitivity, molecular specificity, and imaging resolution capable of detecting submicrometer features in films.^{22–24} The technique has evolved to be an important aspect of surface analysis, and nanometer films have been studied extensively for making measurements such as film composition, uniformity, structural defects, and impurities in thin films ranging from organic semiconductors to biological sensors.^{25–27} However, the analysis of ultrathin and very thin films on inorganic surfaces makes interpretation of the data difficult since the implantation depth of the primary ions would be greater than the film thickness.²⁸ This would generate a collision cascade that overlaps both the organic overlayer and the substrate, complicating the sputtering and ionization processes. Simulations have shown an uplifting of the substrate atoms that increases both sputter yield as well as fragmentation of the overlayer material,^{29–31} but no experimental study has examined these changes in a systematic manner or identified artifacts encountered during ultrathin film analysis, especially for crosslinked plasma polymer films where the sputtering and secondary ion emission processes could vary significantly from molecular films. In this study, plasma polymerized tetraglyme and methane films ranging in thickness from 1 to 20 nm were examined to observe these effects, using primary ion sources of different sizes and energy (Bi_1^+ , $\text{Bi}_3^{+/++}$, Bi_5^{++} , and $\text{C}_{60}^{+/++}$). Tetraglyme is rich in oxygen and included to identify effects of film chemistry on ionization, such as ionization enhancement seen at the silicon oxide interface during depth profiling.^{32,33} Numerous studies have shown the homogeneity, reproducibility, and thickness control of tetraglyme films,^{34–38} and successful depth profiles with very low crosslinking and chemical damage have shown their potential in serving as model systems for characterizing experimental parameters for ToF-SIMS.³⁹ Similarly, extensive literature on plasma polymerized and highly crosslinked methane films exist, with an emphasis on creating reproducible films with great control of optical, thermal, and electrical properties to create hard coatings for many industrial applications.^{40–44}

II. EXPERIMENT

A. Substrate and film preparation

10 cm Si(100) wafers (Silicon Valley Microelectronics, CA) were cut into $1 \times 1 \text{ cm}^2$ square pieces using a Kuliche & Soffa 780 dicing saw with a $15 \mu\text{m}$ diamond impregnated blade (Willow Grove, PA). Distilled water was used to wash away any dicing dust. The silicon pieces were sonicated in dimethyl chloride, acetone, and methanol, twice for 5 min in each solvent. Additionally, they

were soaked in 5% hydrofluoric acid for 5 min to remove the native oxide layer before being rinsed in de-ionized water and used for plasma deposition. The radio-frequency glow discharge (RFGD) deposition system used to produce the crosslinked plasma polymerized films has been described in detail elsewhere.³⁴ The RF used was 13.56 MHz, and reactor walls were heated using electrical heating tapes to prevent monomer condensation. Tetraglyme was preheated to 90 °C to evaporate water molecules, then vaporized at 105–110 °C, and allowed to flow into the reactor at a rate of 1.55 SCCM at 350 mTorr. The glow discharge was maintained at 10 and 20 W for 120 to 600 s to create layers of approximately 1–20 nm thick. The methane monomer was made to flow into the reactor from a pressurized bottle, and deposition was performed at 250 mTorr with a flow rate of 1.55 SCCM. The glow discharge was maintained at 20 and 40 W for 20–480 s to create layers of approximately 2–20 nm thick. For both reactions, the flow valve was shut immediately after stopping the glow discharge to prevent monomer deposition, and the samples were used immediately without any modification or soaking.

B. Film morphology and thickness measurements

Thickness and uniformity of the films were measured using a Dimension 3100 atomic force microscope (Veeco Metrology, Santa Barbara, CA) in intermittent contact mode. Step heights were measured after scratching the film with a piece of silicon, and uniformity was determined from root mean square (rms) roughness within three $50 \times 50 \mu\text{m}^2$ areas per sample. The average roughness was $<0.9 \text{ nm}$ across all film surfaces. Thickness was also measured by fitting the Cauchy model with spectra collected using a Woollam M-2000 spectroscopic ellipsometer. Ellipsometric optical quantities, the phase (Δ) and amplitude (ψ), were acquired from spectra at incidence angles of 65°, 70°, and 75° using wavelengths from 210 to 900 nm. The beam diameter was 1.5 mm. Film thickness and optical constants n (refractive index) and k (extinction coefficient) were then determined iteratively to give a best fit using the Levenberg-Marquardt algorithm. Three locations on each sample were scanned to determine the average thickness of the film.

C. Film chemistry and data analysis

Elemental information was obtained using a Surface Science Instruments S-Probe x-ray photoelectron spectrometer (XPS) equipped with a monochromatized $\text{Al K}\alpha_{1,2}$ source and a 55° photoelectron take-off angle (this was defined as the angle between the sample normal and the axis of the analyzer lens). Survey (analyzer pass energy = 150 eV) and high-resolution C 1s (pass energy = 50 eV) spectra were obtained and analyzed using the Service Physics ESCA 2000 A software. At least three spots on each sample were analyzed. ToF-SIMS was used to obtain chemical information (IONTOF ToF-SIMS 5-100, Münster, Germany). The instrument was equipped with a 25 kV Bi_n^{q+} source ($n = 1, 3$, and 5, $q = 1$ and 2) and a 10 kV C_{60}^{q+} source ($q = 1-2$), both with an incidence angle of 45°. Spectra were acquired over an analysis area of $100 \times 100 \mu\text{m}^2$ (128×128 pixels) with an ion dose of $1 \times 10^{12} \text{ ions/cm}^2$. Prior to each analysis, primary ion currents were measured in a Faraday cup and adjusted so that the signal during the analysis never exceeded 150 000 counts per second to minimize

detector saturation,⁴⁵ but some peaks were still found to be saturated. These peaks were excluded from analysis and spectra were normalized to the sum of selected peaks. The default IONTOF dead-time correction was used when processing the ToF-SIMS data.

Multivariate analysis was performed to gain a better understanding of the data. A separate peak list was created for the methane and tetraglyme samples that included all peaks except known isotopes, saturated peaks, and peaks related to poly (dimethyl siloxane) (PDMS), a common surface contaminant. Since PDMS-related peaks include Si and O containing peaks, this means peaks from the silicon substrate are removed from the analysis. Then, the remaining peaks in each spectrum were normalized to the sum of selected peaks to correct for variations in the secondary ion yields between different spectra. Following normalization, the square root of the data was taken, and then the data were mean centered. Multivariate analysis was performed using principal component analysis (PCA) using the NBTtoolbox (Dan Graham Ph.D., University of Washington) in MATLAB (MathWorks, Inc., Natick, MA), the theory of which is described in detail elsewhere.^{46–49}

III. RESULTS AND DISCUSSION

A. Analysis of plasma-polymerized tetraglyme films

The plots in Fig. 1 show the intensities of representative tetraglyme-related fragments and hydrocarbons as a function of film thickness. A general increase in tetraglyme fragments with film thickness is seen for all primary ion species, indicating a limited number of molecules available for desorption toward the lower end of the film thickness, consistent with simulation results.^{29,30} In this region, nonlinear enhancement from bismuth clusters⁵⁰ or from the impact of large cluster C_{60} ions was not apparent—their total ion yields were less than those obtained using monoatomic bismuth ions, and they generated lower intensities of higher mass peaks. On the other hand, hydrocarbon peak intensities were seen to decrease with film thickness. These hydrocarbons are thought to originate from ambient contamination, and thinner films were expected to show larger relative contributions from hydrocarbon overlayers on both the substrate and tetraglyme. Similar absolute amounts of hydrocarbon contaminants are expected to be present on all films, but as the film thickness increases, the relative amount of hydrocarbon contaminants will decrease. The hydrocarbon intensities also seem to depend on primary ion size. Smaller primary ions such as Bi_1^+ generated the highest intensity of larger tetraglyme fragments, presumably due to reduced occurrences of collision-induced dissociation as related to how energy is deposited for small versus larger ions.^{29,30} Conversely, higher intensities of hydrocarbons are seen for larger primary ions likely due to increased fragmentation.

PCA was carried out using all spectra from the tetraglyme films using all primary ion sources. Two general trends can be seen in the PC1 scores plot. First, it can be seen that overall, the score values decrease with increasing primary ion size (Bi_1^+ scores > Bi_3^+ scores > Bi_5^{++} scores > C_{60}^+ scores > C_{60}^{++} scores). Then, within the data for a given primary ion, the score values increase with increasing film thickness. When interpreting PCA results, in general, peaks that show a higher loading value on a given side of a PC loadings axis will show a higher relative intensity

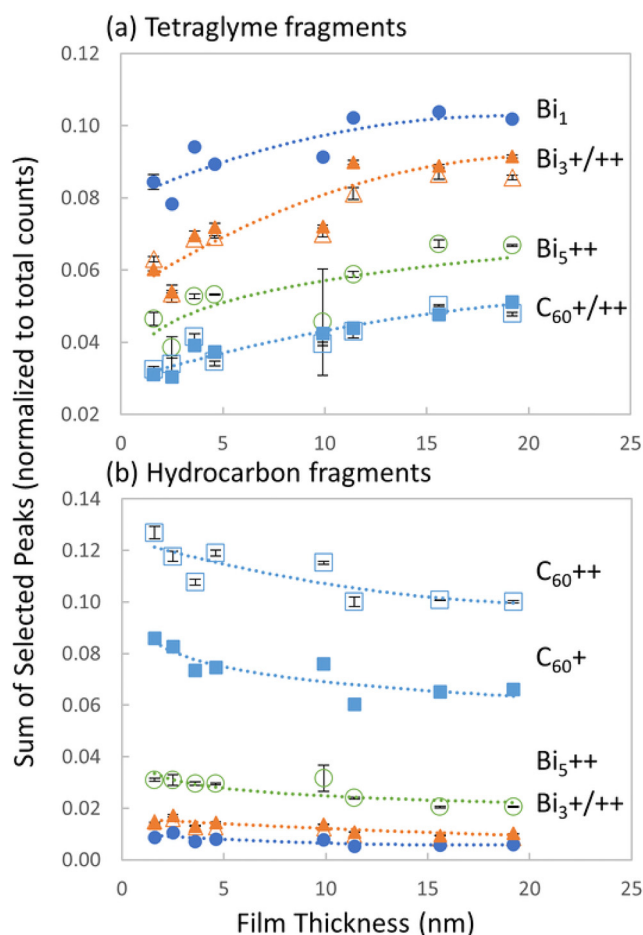


FIG. 1. Integrated ToF-SIMS intensities of the top ten peaks in Table I showing the (a) tetraglyme-related peaks and (b) hydrocarbon peaks as a function of film thickness, using Bi_1^+ (●), Bi_3^+ (▲), Bi_5^{++} (△), Bi_5^{++} (○), C_{60}^+ (■), and C_{60}^{++} (□) primary ions. The peaks were normalized to the total counts. All data points represent an average of at least three measurements, and error bars were based on standard deviation. Data for films less than 2 nm in thickness were not included due to lack of film integrity (i.e., island formation). The dotted lines in the figures are trend lines for each set of points.

in samples with a higher score value on the same side of the PC scores axis.^{48,49} This means that peaks with positive loadings will, in general, have higher relative intensities in samples with positive scores and that peaks with negative loadings will, in general, have higher relative intensities in samples with negative scores.^{48,49}

The positive loadings show a series of $C_xH_yO_z$ peaks that are related to the tetraglyme structure, while the negative loadings show a series of hydrocarbons of the formula C_xH_y (where typically $x \geq y$) along with Na and K presumably from the substrate [see Fig. 2(b) and Table I]. Data shown in Fig. 2 suggests that spectra with positive scores (Bi_1^+ , Bi_3^+ , Bi_5^{++}) will show a higher relative intensity of tetraglyme peaks (positive loadings), while spectra with negative scores (C_{60}^+ and C_{60}^{++}) will show a higher relative

TABLE I. Highest positive and negative PC1 loadings from the tetraglyme data. m/z , centroid mass; PC01, loading value on PC1; Labels, most likely identity based on the centroid mass. Consistent with chemical structure suggested by Menzies *et al.* (Ref. 51).

m/z	PC01	Labels
103.0821	0.2093	C5H1102
71.0526	0.1954	C4H70
87.0471	0.1854	C4H702
73.0692	0.1752	C4H90
73.0324	0.1571	C3H502
89.0626	0.1389	C4H902
85.0677	0.1388	C5H90
101.0607	0.1244	C5H902
69.0363	0.1227	C4H50
75.0486	0.1175	C3H702
99.0458	0.116	C5H702
58.0402	0.1059	C3H60
83.0531	0.0943	C5H70
115.0808	0.0898	C6H1102
57.0334	0.0857	C3H50
113.065	0.0838	C6H902
72.0589	0.0788	C4H80
85.0X6	0.0783	C4H502
97.0658	0.0754	C6H90
99.0805	0.0667	C6H110
87.0817	0.0657	C5H110
95.0503	0.0617	C6H70
109.0672	0.0596	C7H90
61.0281	0.0578	C2H502
84.0585	0.0572	C5H80
125.0668	0.0563	C7H902
111.0835	0.0547	C7H110
111.0474	0.0542	C6H702
39.0215	-0.3289	C3H3
50.0124	-0.264	C4H2
38.0124	-0.2439	C3H2 or CH3Na
51.021	-0.2249	C4H3
62.012	-0.1971	C5H2 or C3H3Na
37.0045	-0.179	CH2Na
26.0137	-0.1756	C2H2
29.0019	-0.165	CHO
74.0147	-0.1581	C6H2
61.006	-0.1419	C5H or C3H2Na
86.0139	-0.1064	C7H2
22.9901	-0.0974	Na
98.0102	-0.0931	C8H2
28.0303	-0.0853	C2H4
31.0185	-0.0831	CH30
42.0083	-0.0827	C2H20
52.0286	-0.0824	C4H4
29.0396	-0.0816	C2H5
38.9631	-0.0809	K
30.0341	-0.0801	CH4N
75.0246	-0.08	C6H3
40.0291	-0.0792	C3H4

TABLE I. (Continued.)

m/z	PC01	Labels
19.0189	-0.0735	H30
52.9994	-0.0619	C3HO
25.0056	-0.0611	C2H
110.0105	-0.0539	C9H2
64.0287	-0.05	C5H4
53.0378	-0.0499	C4H5

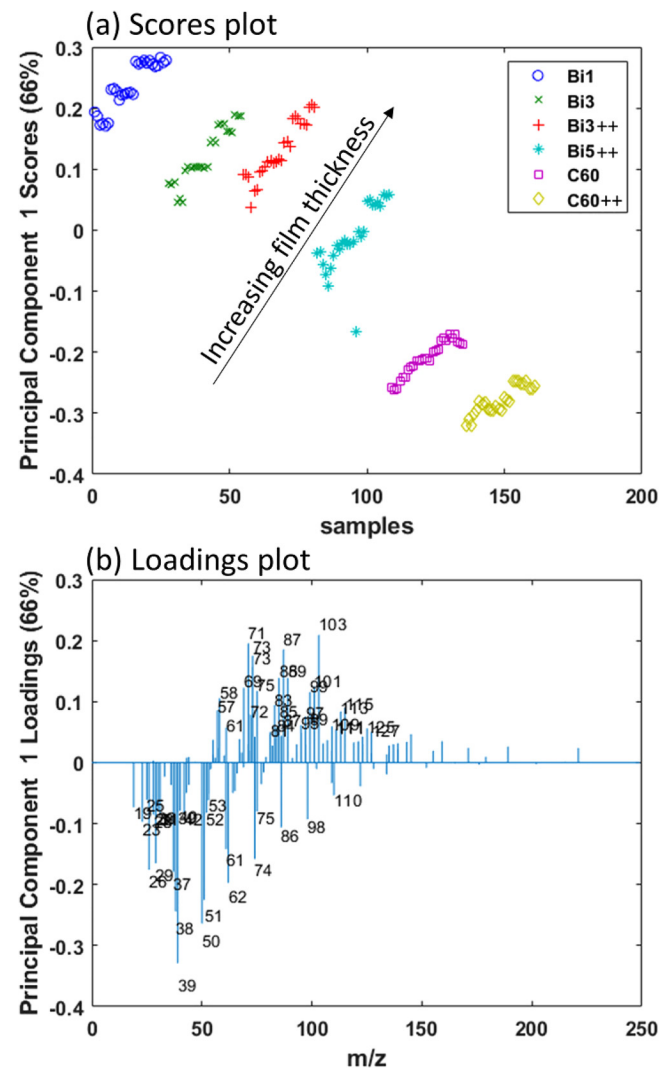


FIG. 2. (a) PC1 scores plot constructed from PCA of the secondary ion mass spectra from tetraglyme RFGD-deposited films, showing separation of samples based on the primary ion used and the thickness of the sample. (b) PC1 loadings plot showing the peaks responsible for the separation seen in the scores plot.

02 September 2025 17:23:29

intensity of hydrocarbon peaks. See the text and figures in the supporting material for examples from the raw data that demonstrate these trends.⁵²

Consistent with data from Fig. 1, the use of bismuth primary ions corresponds with a higher yield of tetraglyme fragments, while the use of C_{60} primary ions is seen to generate a higher yield of hydrocarbon fragments. These hydrocarbons likely originate from ambient contamination, given that their relative intensity increases for thinner films. It is interesting to note that the samples analyzed by C_{60} all showed negative loadings, suggesting that the relative intensity of C_xH_y peaks is higher when analyzed by C_{60} . As mentioned above, this could be related to the increased fragmentation of the target molecule or the formation of carbon adducts from the dissociated fragments of the primary ion, possibly similar in mechanism to the formation of deuterated secondary ions from the impact of deuterated water cluster sources.⁵³ This could also explain the major hydrocarbon peaks in the negative loadings being separated by the mass of carbon and not methylene groups, which is typical for the fragmentation of polymers (i.e., separation by $m/z = 12$, where the peaks occur at $m/z = 50, 62, 74, 86, 98, 110$, etc.).

These results were correlated with composition measurements made by XPS. As can be seen in Fig. 3, the attenuation of the Si 2p signal correlates well with film thickness as does the increase in C 1s. This is consistent with SIMS results in Fig. 1(a) where the increase in the intensity of tetraglyme-related fragments is proportional to the amount of material that can be desorbed from the surface. Oxygen content remains constant, but the inspection of the C–O line suggests slight changes in plasma-polymerized chemistry based on the glow discharge power, where 10 W was used mainly for the thinnest films and 20 W for thin films. This small change in chemistry was not captured in the SIMS results, which

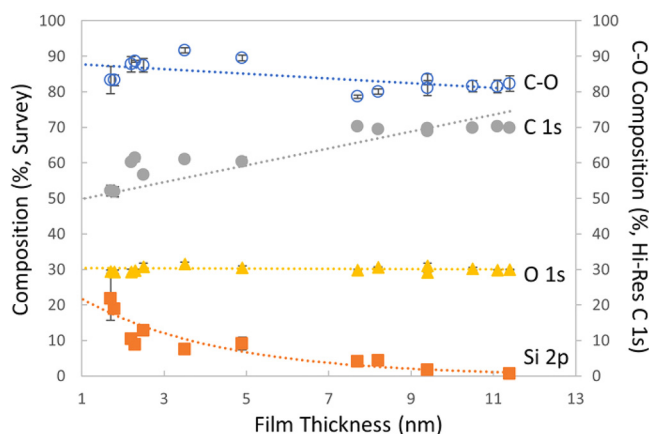


FIG. 3. XPS elemental analysis of the tetraglyme plasma-deposited films, showing the C 1s (●), O 1s (▲), Si 2p (■), and high-resolution C–O (○) lines as a function of thickness. All data points represent an average of at least three measurements, and error bars were based on standard deviation. A control (glow discharge without monomers) was not used since the plasma process can generate oxides from residual gases that do not resemble the chemistry of the monomer. The dotted lines in the figures are trend lines for each set of points.

either suggests that the ionized fragments are small enough to not reflect the larger structural changes or the C–O lines are affected by the presence of ambient carbon contamination whose contributions can be relatively large for these ultrathin films.

In simulations of molecular thin film sputtering, one interesting aspect that is discussed is the uplifting of organic molecules by substrate particles in a “catapultlike action.”⁵⁴ This phenomenon was not observed in the current study, which is not surprising since RFGD thin films are cross-linked and should have a different emission process than molecular thin films. The structure of the film being analyzed can affect the sputtering and emission process. For example, Bi_1^+ has been shown to be more surface sensitive than Bi_3^+ when analyzing polyurethane block copolymers even though the penetration depth of Bi_1^+ is deeper than Bi_3^+ .⁵⁵ Transport of ions in matter (TRIM) simulations of organic overlayers suggests that for a Bi_1^+ primary ion with 25keV energy and 45° incidence angle, the sputtering of silicon substrate particles can occur for films as thick as 5 nm (silicon comprises 0.2% of the sputtered

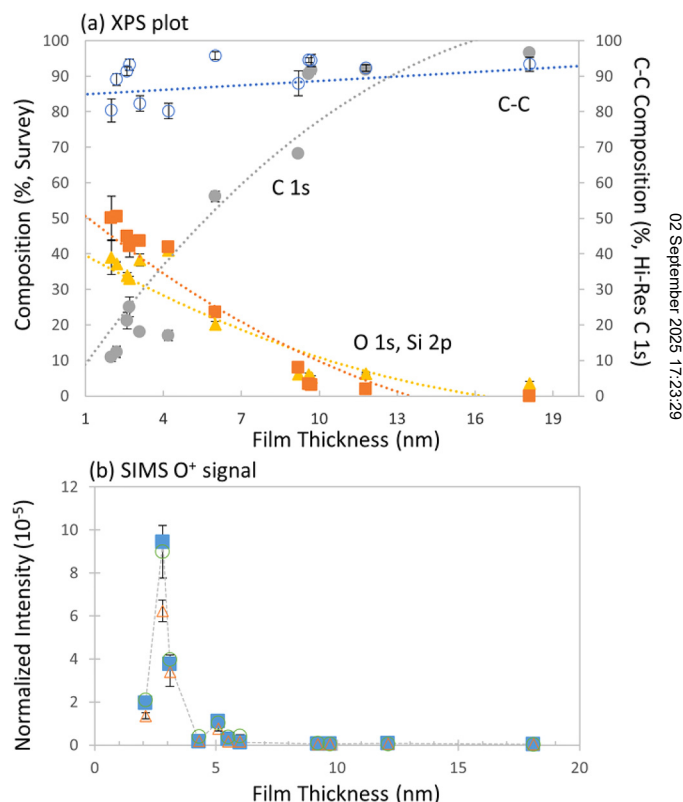


FIG. 4. (a) XPS elemental analysis of the methane plasma deposited films, showing the C 1s (●), O 1s (▲), Si 2p (■), and high-resolution C–C (○) lines as a function of thickness. (b) ToF-SIMS intensities of oxygen normalized to total counts as a function of film thickness. Data obtained using Bi_3^{++} (Δ), Bi_5^{++} (○), and C_{60}^+ (■) primary ions are shown, but all ion sources displayed the same behavior. All data points represent an average of at least three measurements, and error bars were based on standard deviation. The dotted lines in the figures are trend lines for each set of points.

atoms). Naturally, the use of bismuth clusters and C_{60} ions would sputter a greater amount of substrate particles owing to the formation of a wider crater,²⁸ but the lack of any yield enhancement suggests that if there are any uplifted molecules they are mostly neutral in nature. Also, the plasma polymerization process occurs at pressures of a few hundred mTorr, which results in surface oxidation from the integration of oxygen from residual air and water vapor in the glow discharge chamber into the substrate (discussed later). Typically, matrix-related yield enhancements are seen in depth profiles at the film–silicon oxide interface,^{32,33} but no such observations were made in the current study.

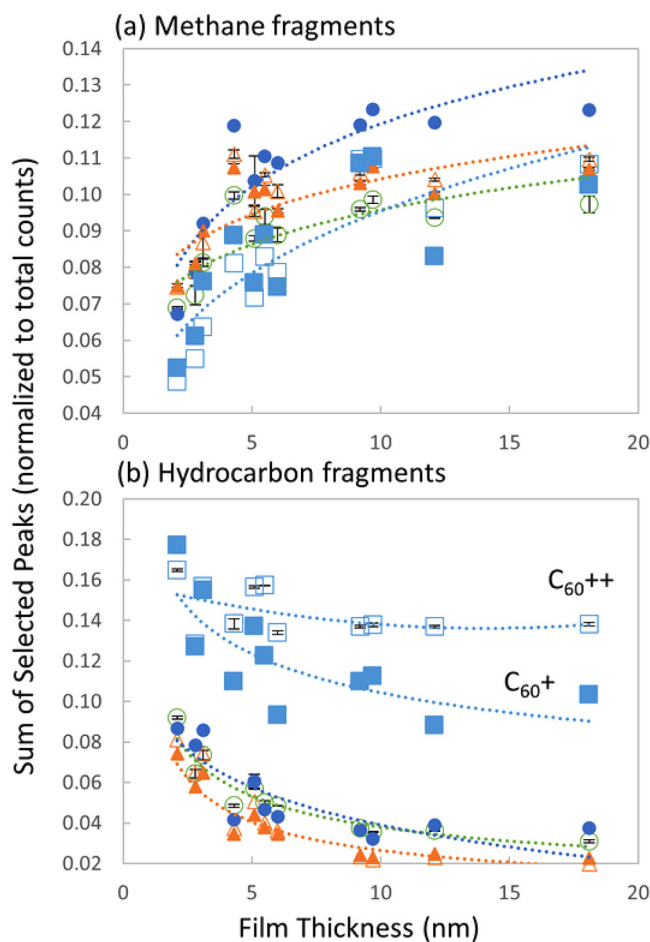


FIG. 5. Integrated ToF-SIMS intensities of the top ten peaks in Table II showing the (a) polymerized methane-related peaks and (b) hydrocarbon contamination peaks as a function of film thickness, using Bi_1^+ (●), Bi_3^+ (▲), Bi_3^{++} (Δ), Bi_5^{++} (○), C_{60}^+ (●), and C_{60}^{++} (□) primary ions. The peaks were normalized to the total counts. All data points represent an average of at least three measurements, and error bars were based on standard deviation. The dotted lines in the figures are trend lines for each set of points.

TABLE II. Highest positive and negative PC1 loadings from the polymerized methane data. m/z , centroid mass; PC01, loading value on PC1; Labels, most likely identity based on the centroid mass.

m/z	PC01	Labels
69.0728	0.179	C5H9
81.0741	0.15	C6H9
95.0868	0.1184	C7H11
105.0691	0.1171	C8H9
165.0578	0.1109	C9H9O3
57.07	0.1084	C4H9
119.0899	0.1076	C9H11 possible
67.0557	0.1054	C5H7
43.0547	0.1044	C3H7
109.104	0.0965	C8H13
83.0896	0.0943	C6H11
93.0706	0.0935	C7H9
141.0628	0.0934	C10H7N or C11H9
179.0727	0.0903	C10H11O3
131.0863	0.0885	C10H11
145.0993	0.0874	C11H13
155.0781	0.0859	C12H11 or C11H9N or C8H11O3
143.0805	0.0831	C11H11
215.0616	0.0816	C13H11O3
202.0476	0.0815	C12H10O3
91.0538	0.0812	C7H7
107.0867	0.0803	C8H11
129.0679	0.078	C10H9
157.0967	0.0758	C12H13
133.1016	0.0753	C10H13
121.1075	0.0747	C9H13
142.0712	0.0738	C11H10 or C10H8N
191.0658	0.0734	C11H11O3
189.0501	0.0716	C11H9O3
169.093	0.0712	C13H13 or C9H13O3
193.0887	0.0706	C11H13O3 or C15H13
39.0222	−0.3128	C3H3
27.0233	−0.2949	C2H3
50.0135	−0.1954	C4H2
51.0224	−0.179	C4H3
38.0137	−0.1779	C3H2
62.013	−0.1396	C5H2
26.0148	−0.1374	C2H2
63.0217	−0.1353	C5H3
37.0057	−0.1349	C3H
42.9981	−0.1345	C2F
30.0344	−0.1339	CH4N
29.0018	−0.13	CHO
74.0171	−0.127	C6H2
15.0234	−0.1209	CH3
40.0296	−0.102	C3H4
29.0393	−0.0969	C2H5
42.0342	−0.093	C2H4N
52.0299	−0.0928	C4H4
31.0182	−0.0875	CH3O
75.0288	−0.0836	C6H3

02 September 2025 17:23:29

TABLE II. (Continued.)

m/z	PC01	Labels
86.0148	−0.0828	C7H2
49.0037	−0.081	C4H possible
85.0108	−0.0789	C7H
98.0116	−0.0701	C8H2
42.0089	−0.0643	C2H2O
11.9991	−0.0607	C
73.0116	−0.0599	C6H
56.0485	−0.0597	C3H6N
22.9896	−0.0576	Na
45.0333	−0.0551	C2H5O
13.0073	−0.0547	CH

B. Analysis of plasma polymerization methane films

To isolate the effects of oxygen within the organic overlayer on secondary ion yield, plasma-polymerized methane films were chosen as a comparison. Verification of the absence of oxygen in the film was done using both XPS and SIMS, as shown in Fig. 4. The concentration of oxygen in the substrate, as indicated by O 1s and indirectly through Si 2p, shows complete attenuation as the film gets thicker. However, due to the lower average atomic density of the film and its subsequent effect on inelastic electron scattering, full attenuation does not occur until the film thickness reaches around 12 nm. SIMS is not as sensitive to small differences in atomic density and is able to show that full oxygen attenuation occurs around 5 nm thickness, which is the typical information depth for organic films.²⁸ Also, as discussed previously, the plasma polymerization process leads to surface oxidation due to the integration of oxygen from residual air and water vapor in the chamber. The concerted change in composition and intensity of the oxygen and silicon signals in the XPS and SIMS data, respectively, shows that the substrate does become partially oxidized in this process.

SIMS data showing the integrated intensities of fragments from the methane thin films are shown in Fig. 5. Similar to the tetraglyme thin films, a general increase in polymerized methane fragments with increasing film thickness is seen for all primary ion species. Again, secondary ion yields of these fragments were highest when using the Bi₁⁺ primary ion, and nonlinear yield enhancements were not seen from bismuth clusters or larger C₆₀ ions. Additionally, despite the presence of an oxidized layer at the film–substrate interface, no matrix-related ionization enhancement was observed. Perhaps the results may already reflect a matrix-enhanced ionization or the phenomenon may require exposing a pristine, higher energy layer in vacuum during depth profiling. The data also showed higher intensities of ambient carbon contamination dominating the region of thinner film thickness, which can also be seen in the XPS data where the large scatter in the C–C composition indirectly suggests the presence of nonfilm-related material.

Interestingly, there was a disparity in the intensities of fragments generated using C₆₀ depending on their source. Peaks of methane plasma polymer fragments were a factor of two lower

compared to peaks originating from ambient carbon contamination. Given the methane plasma polymer's highly crosslinked nature⁵⁶ and its use as a protection layer⁵⁷ for materials from the environment, these diamondlike films⁵⁸ likely have a much higher degree of cross-linking compared to typical polymer films and lead to lower sputtered volumes for large clusters such as C₆₀ where the energy per nucleon (eV/n) of the C₆₀ source is much lower than that of bismuth. The effect of cross-linking on sputter yield is discussed in detail by Seah *et al.*,⁵⁹ where the sputter yield is inversely related to the degree of cross-linking.

Figure 6 shows the PCA results of SIMS data obtained from the methane plasma polymer. PC1 did not show clear differences in fragmentation patterns among primary ion sources, but it did

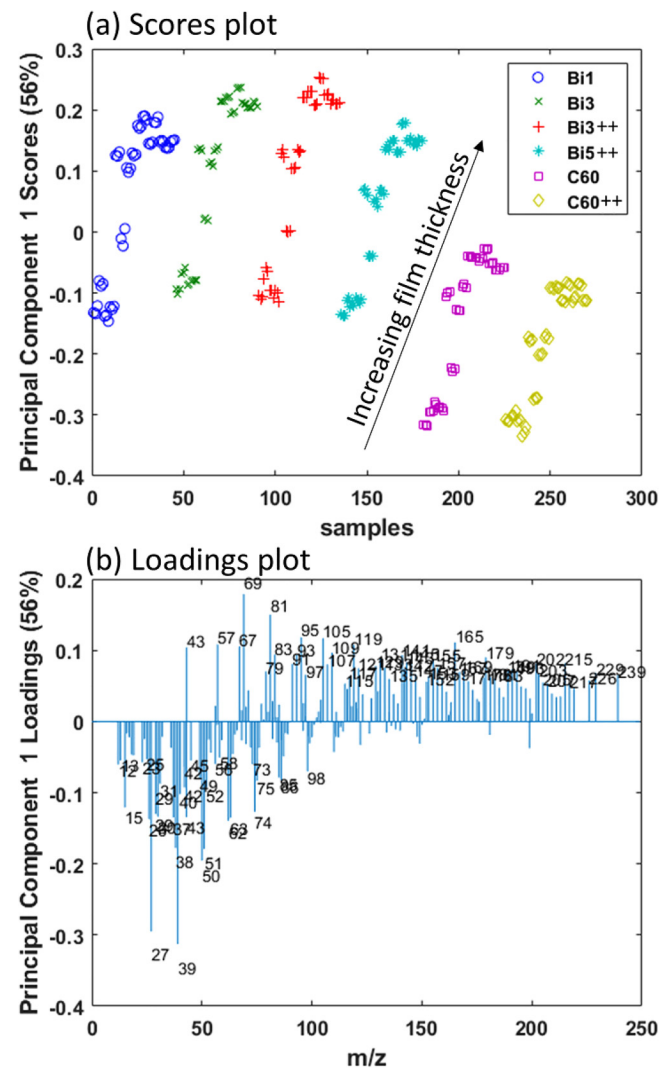


FIG. 6. (a) PC1 scores plot from RFGD-deposited methane films, showing the separation by film thickness for different primary ions. (b) PC1 loadings plot showing the peaks responsible for the separation seen in the scores plot.

02 September 2025 17:23:29

show a general trend of increasing film thickness with increasing score value. The positive loadings showed higher mass hydrocarbon fragments, whereas the negative loadings showed shorter hydrocarbons. The increased intensity of larger C_xH_y ions with thicker films is consistent with the increased presence of highly crosslinked chains resulting from the plasma polymerization. On the other hand, thinner films showed hydrocarbon peaks similar to those seen in the tetraglyme data with similar peak distributions along with a similar mass range from about m/z 27 to 110.

IV. CONCLUSIONS

This study examined the effect of film thickness on the ionization of the sputtered flux. At ultrathin film thickness below 10 nm, these crosslinked plasma polymerized films deposited onto a silicon wafer showed characteristics that are not usually observed for bulk films with lower degrees of cross-linking. (1) The presence of ambient carbon contamination is proportionately very large and tended to dominate both the mass spectra and PCA results. Trying to discriminate the film fragments from contaminants, especially for films like highly crosslinked materials such as plasma-deposited methane films that do not contain oxygen groups, may prove difficult but PCA can help. (2) At these film thicknesses, the effect of cluster-induced nonlinear enhancement was not observed. This is reasonable since clusters have higher sputter yields from the formation of large craters. Ultrathin films do not provide the necessary volume for enhancement, which agreed well with simulations reported in literature. (3) The simulations also suggested that fragments of thin films experience a catapulting action from the uplifting of the substrate atoms, thereby increasing the sputter yield, but this was not seen to translate into higher secondary ion yields. (4) The extent of fragmentation was higher (i.e., smaller fragments) for thinner films, suggesting that the kinetic energy of the primary ion remains more confined to the organic overlayer. This may be caused by a subset of collision events induced by the recoiling substrate atoms. (5) Finally, these ultrathin films did not display yield enhancements that are typically observed at film-silicon interfaces during depth profiles, hinting that chemical and/or topographical changes at the crater bottom during depth profiling may be responsible for the phenomenon. It is noted that these results are likely specific to the crosslinked plasma polymers chosen for this study. Further research would be required to know whether these trends are consistent across multiple materials with varying degrees of cross-linking.

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AUTHOR DECLARATIONS

Conflict of interest

The authors have no conflicts to disclose.

Author Contributions

Shin Muramoto: Conceptualization (lead); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Writing – original draft (lead); Writing – review & editing (equal). **Daniel J. Graham:** Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (lead); Writing – original draft (equal); Writing – review & editing (equal). **David G. Castner:** Funding acquisition (lead); Project administration (lead); Resources (equal); Supervision (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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02 September 2025 17:23:29

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